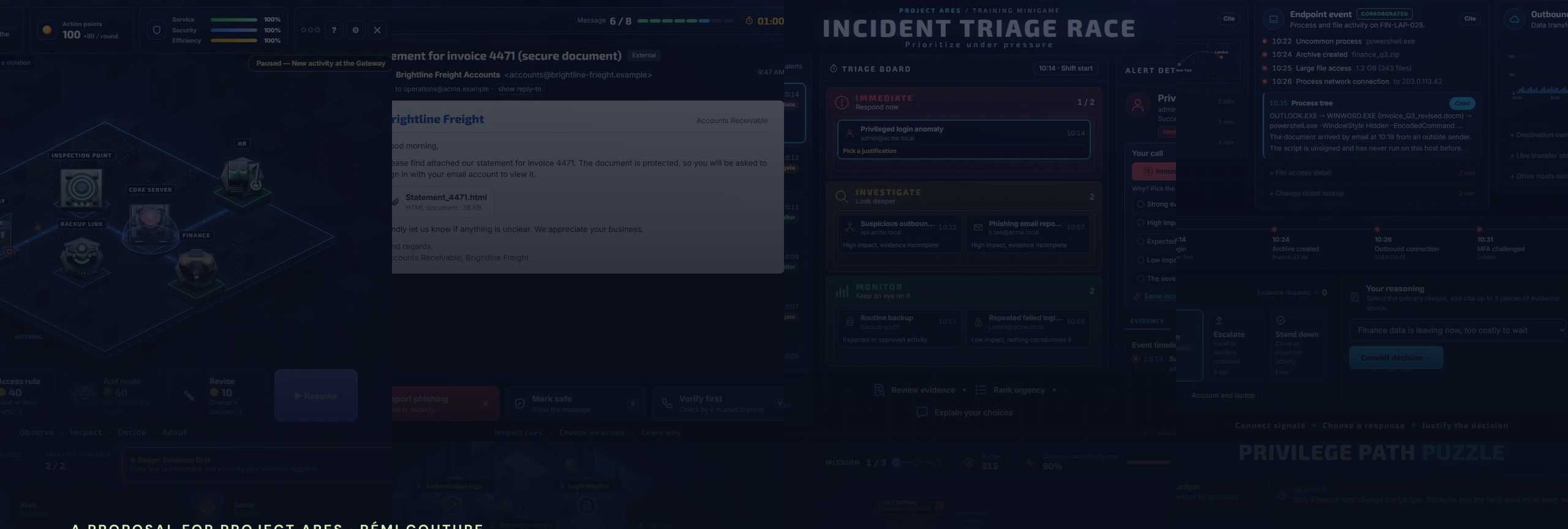



A PROPOSAL FOR PROJECT ARES · RÉMI COUTURE

# Twenty-two training games, already playable.

What I have built for Ares on my own initiative, why it teaches, how it fits the platform, and a plan to keep building games for Circadence and its clients.

THE TRAILER

# All 22 games in seventy seconds.



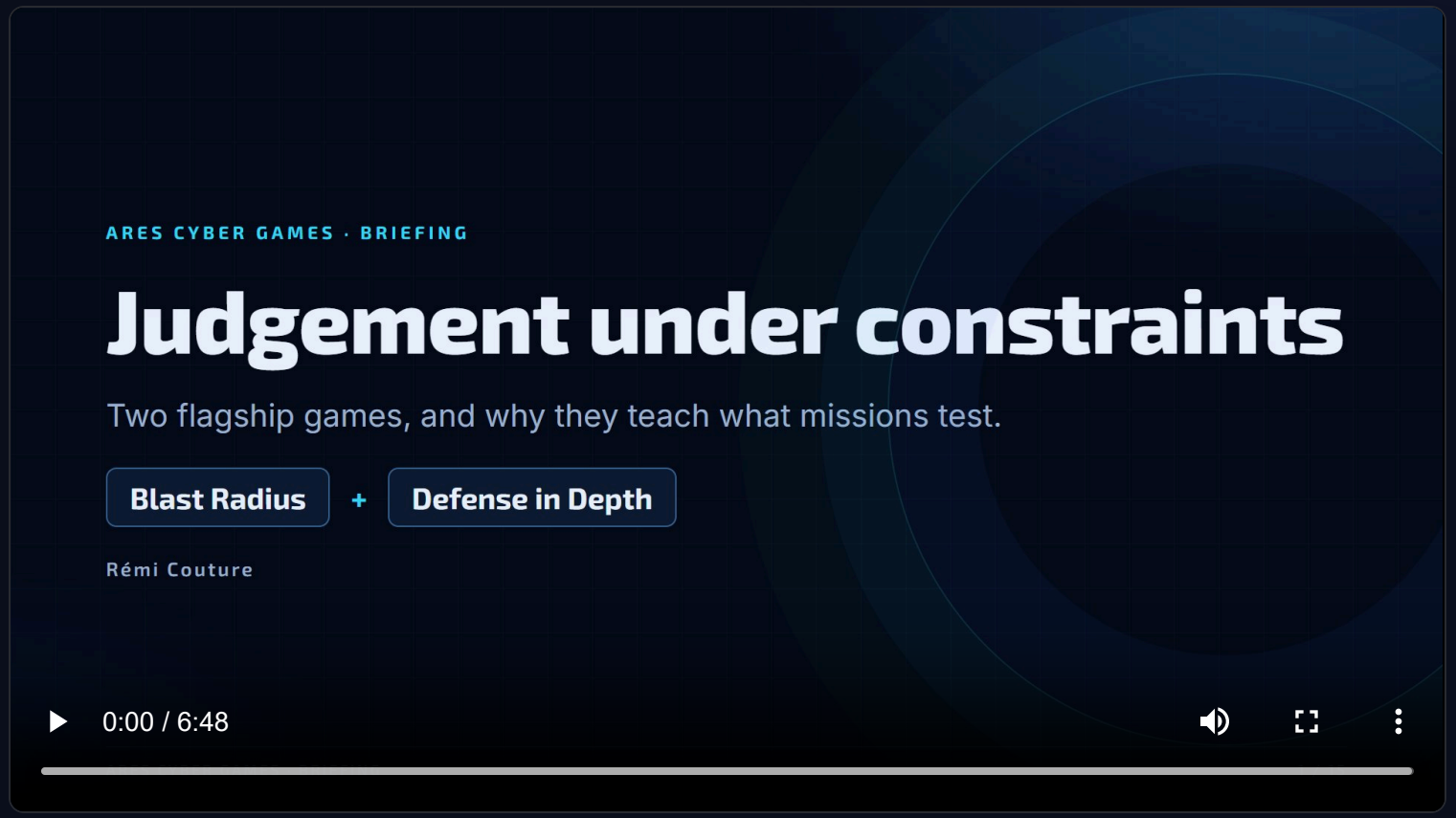

## What is in it

- The original eight, profiled in this proposal.
- Eight more built since, from incident scoping to firewall rule order.
- Six remixes of the Games Hub titles, with new twists.

## Play them all

remiclassic.github.io/ares-cyber-games: every game, each with a how-to-play video. Real gameplay, an original soundtrack, cut to the beat.

# The two flagship games, explained in seven minutes.




### Blast Radius

Scoping before containment: inference from aggregate telemetry, graded on whether the evidence proved the answer.

### Defense in Depth

Resilience against an adaptive attacker: every stop is re-run without the control that made it.

### Slides and notes

ares-cyber-games/briefing.html: the deck, with speaker notes for each slide.

THE SHORT VERSION

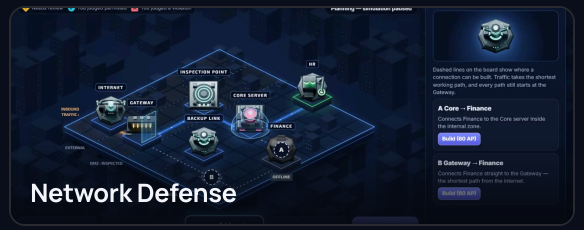
# I have twenty-two working educational games for Ares. I would like to keep making them.

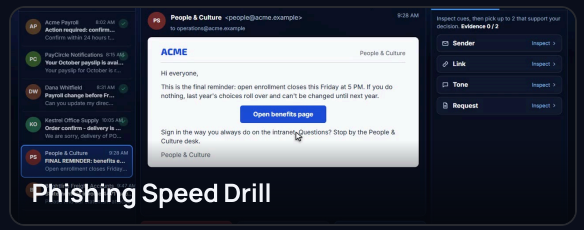
**22**
games, each playable from briefing to debrief in a browser

**318**
automated tests, all passing, covering the learning rules of every game

**21**
different genres, from isometric strategy and SQL hunts to a card duel, a physics sticker pile and an orbital shooter

**0**
ranges to spin up. Each game is a static bundle that embeds in a course page

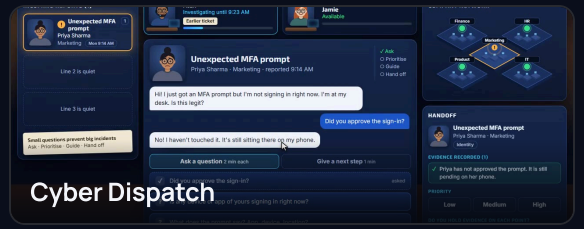


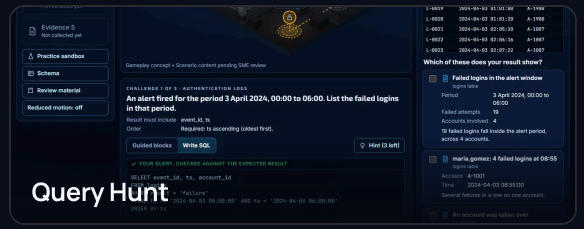
















## What this is

Real training content, not mock-ups: authored scenarios, evidence, policies, queries, and a written explanation for every important mistake. The next pages quote it.

## Why it matters to Ares

Missions need a live range. These need nothing. They give learners practice between missions, give sales a demo that runs anywhere, and fill slots the platform has already reserved for games.

## What I am asking

Keep me on to take these from prototype to production with the games and content teams, and to keep building new ones. I have set out three ways to do that, including a half-time option.

THE ORIGINAL EIGHT

# One skill each. Three to five minutes. A debrief that explains the result.

| GAME | PLAYS LIKE | WHAT THE LEARNER PRACTISES | AUTHORED CONTENT |
|---|---|---|---|
| **Network Defense**<br>Keep Finance Online | Isometric strategy | Routes and dependencies, reading evidence, writing a scoped access rule, backup routes that keep policy | 3 rounds, 3 traffic flows, 10 evidence lines, 4 policy lines, 2 reasoning checkpoints, a final no-hint challenge |
| **Phishing Speed Drill**<br>The Payroll Request | Timed inbox | Sender, reply-to and real link destination; when to report, mark safe, or verify by phone | 8 messages (3 phishing, 3 legitimate, 2 need verification), 32 cues, each with a reason |
| **Incident Triage Race**<br>The Busy Morning | Alert board | Prioritising by evidence and business impact, not the severity tag; revising when facts change | 6 alerts, 29 timeline evidence items, 18 authored assessments, 2 mid-shift updates |
| **SOC Dashboard Challenge**<br>A Strange Login | Investigation case | Telling discriminating evidence from evidence that fits both stories; proportionate response under time | 2 hidden truths with identical openings, 9 evidence items in 2 variants, 11 timed events |
| **Cyber Dispatch**<br>First Shift | Dispatcher desk | Asking the right questions, giving the safe next step in time, handing off only what is confirmed | 6 reports across 2 shifts, 38 questions, 26 facts, 24 guidance options |
| **Query Hunt**<br>Unusual Access | SQL investigation | WHERE, GROUP BY, ORDER BY, JOIN on real tables, and not over-claiming from the results | 5 challenges, 4 tables, 2 datasets, real SQLite running in the browser |
| **Cloud Command**<br>First Orbit | Arcade flight | Capacity through a dependency, cost against availability, least privilege, the IaaS responsibility split | 3 checkpoints, 11 evidence items, 12 justifications with feedback, 7 permissions |
| **Privilege Path Puzzle**<br>Who Can Read Payroll | Graph puzzle | Tracing nested groups and inherited grants, and making the smallest fix that keeps business access | 3 puzzles, 22 nodes, 13 access checks each with a business reason |

EIGHT MORE, BUILT SINCE

# Same rules: one skill each, every mistake explained, lazy strategies proven to lose.

| GAME | PLAYS LIKE | WHAT THE LEARNER PRACTISES | AUTHORED CONTENT |
|---|---|---|---|
| **Secure Stack** | Physics sticker pile | Matching each threat to the control that really stops it; wrong drops explain why | 5 islands, 17 threats and defences, a reason for every pairing, a field guide |
| **Spot the Phish** | Inbox with a coach | Reading an inbox: report, verify another way, or continue safely | 10 messages, then 4 transfer messages; hosted on its own site |
| **Blast Radius** | Deduction puzzle | Scoping a compromise from aggregate sensors before containing it | 3 cases with 2 truths each, every case proven solvable within its triage budget |
| **Defense in Depth** | Defence planning board | Layering controls across the attack chain while attackers adapt | 3 waves, 16 ATT&CK techniques, 15 controls with cost and friction |
| **Counterplay** | Card duel | Reading raw telemetry, naming the technique, playing the response that removes it | 3 incidents, 16 techniques with look-alikes, 22 response cards |
| **Bit by Bit** | Gear workbench | Binary, subnet masks, sizing a subnet, finding the network an alert came from | 10 tasks, every wrong answer explained |
| **Rule Order** | Rule-ordering puzzle | First-match firewall logic, shadowed rules, default action, least privilege | 4 change tickets with test packets; the fewest changes found by search |
| **Spot the Risks** | Hidden-object scene | Physical and social-engineering risks at the office and at home, with decoys | 2 illustrated scenes, 13 risks and 4 decoys, an issue and fix for each |

SIX REMIXES OF THE GAMES HUB

# The hub games' topics, rebuilt around a judgment twist.

| REMIX | REMIX OF | THE TWIST |
|---|---|---|
| **Pattern Patrol** | RegExile (games team) | Regex as detection engineering: catch the attack, not the noise |
| **Port Authority** | Portflow (games team) | Ports by policy: allow, block or upgrade cleartext |
| **Confidence Board** | Trivialoot (games team) | A six-category board with confidence wagers that reward calibration |
| **Chain Breaker** | Cylitaire: Kill Chain (games team) | Rebuild an incident on the kill chain, then break it earliest |
| **Tactic Orbit** | Astro-Defense (games team) | Tag incoming techniques with their tactic to raise the shield |
| **Blind Spots** | Threat Modeler (games team) | Place sensors on a budget; scored on how early each attack path is seen |

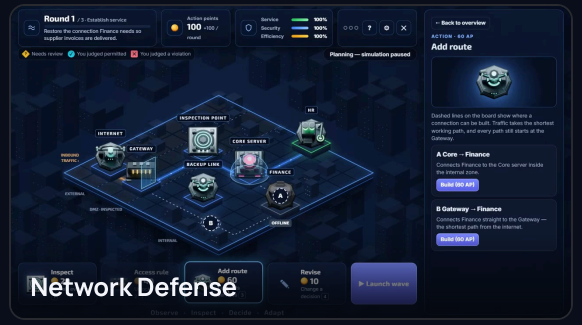
## Why remix

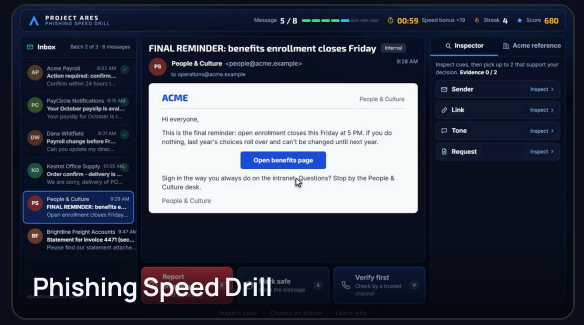
The hub games teach recall of frameworks well. Each remix keeps the topic and asks for a decision on top: a detection that must not over-fire, a confidence you must own, a chain you must break early.

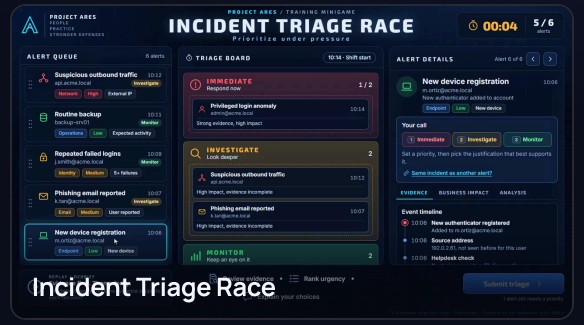
## How they were made

From each game's topic and key art only. No code, levels, text or art copied, and every remix credits its original. They are an offer to the games team, not a replacement.

# Every clip is real play: a decision, a mistake, the explanation, the fix.

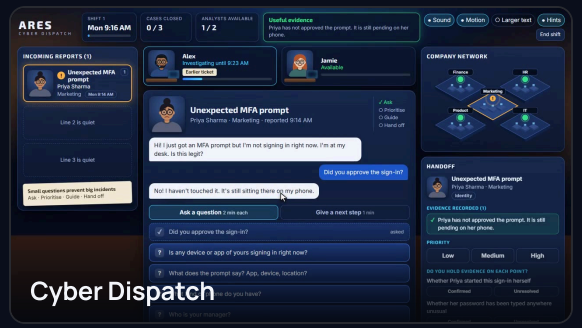


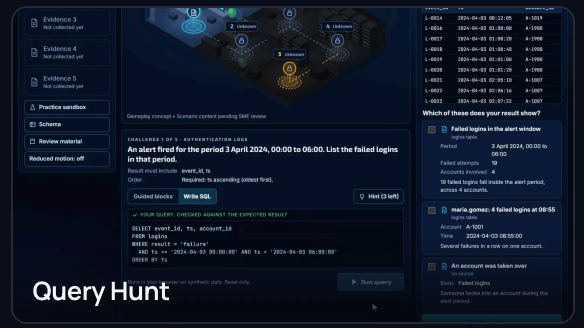





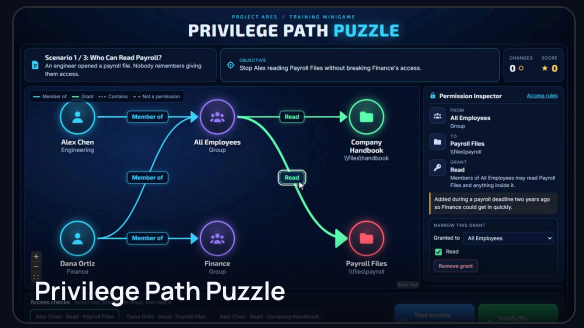




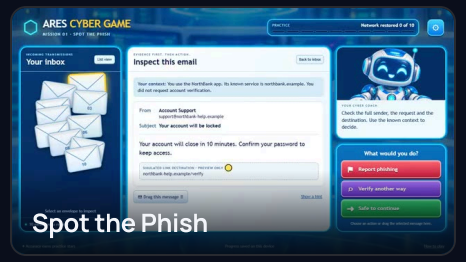






# The newer games and the remixes, recorded from the working builds.

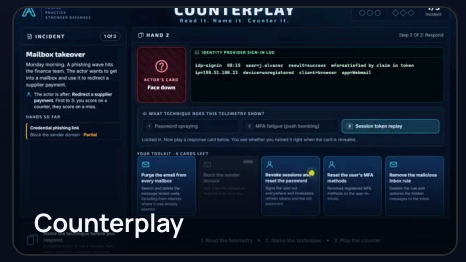




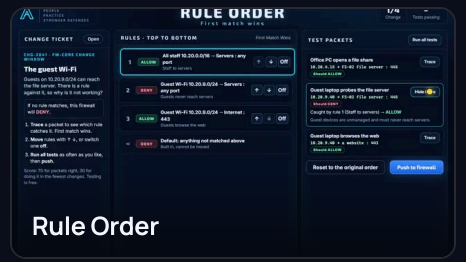


















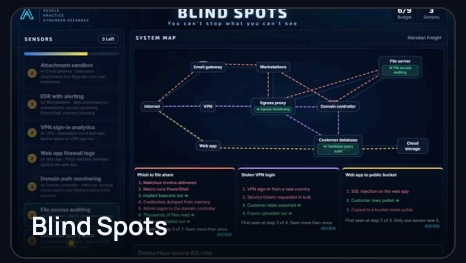




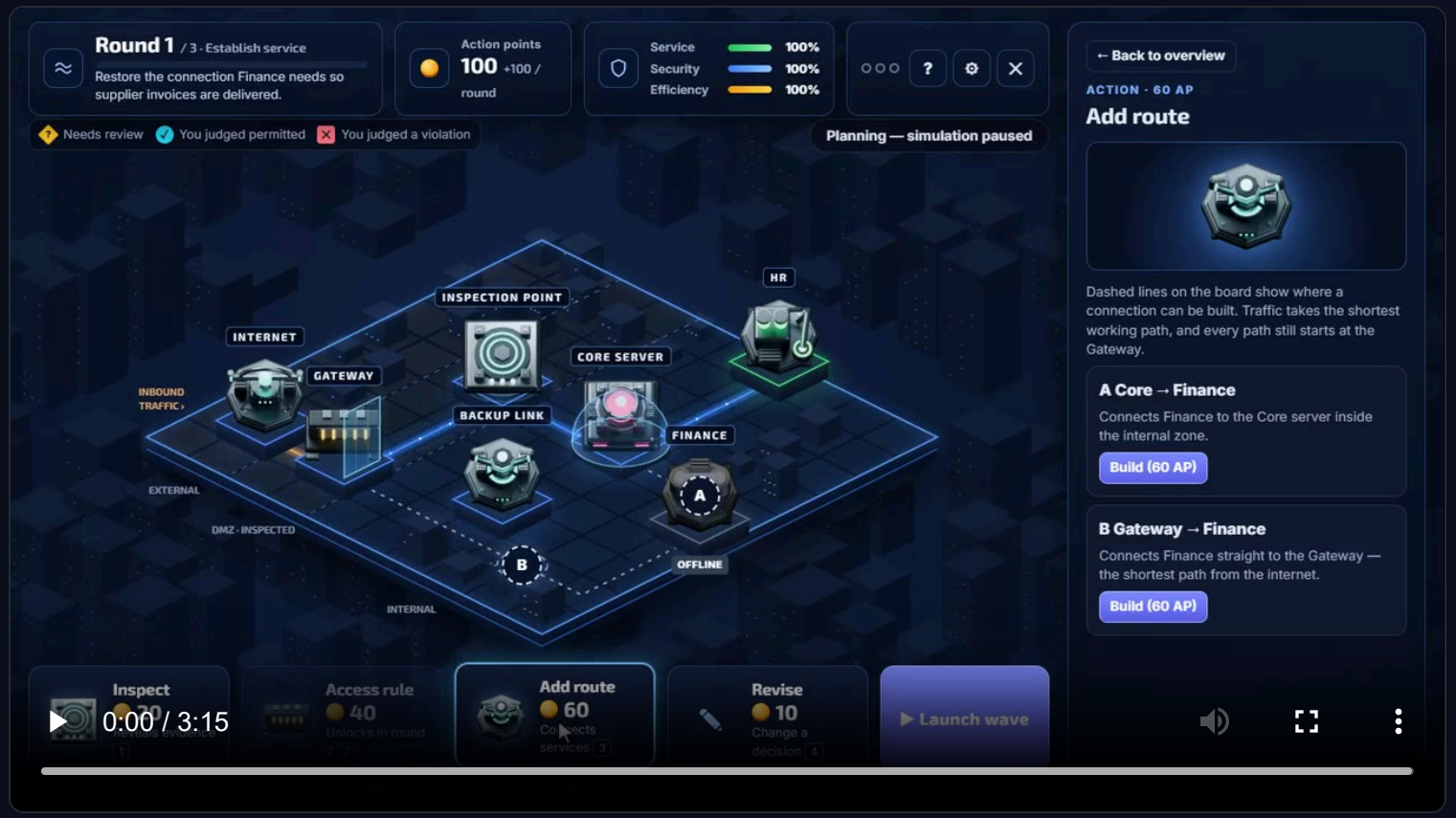






# The gameplay reel: every game, one mistake, one explanation, one fix.



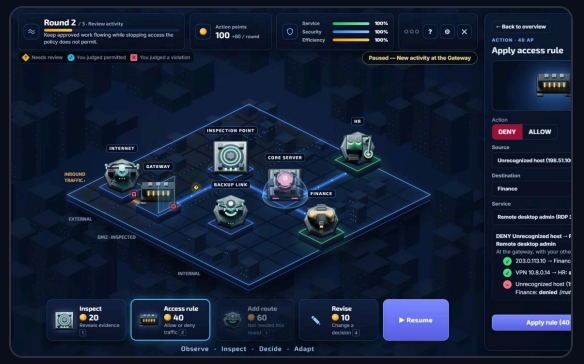

## What to look for

- The learner makes a real decision with real content.
- The game explains a mistake in one or two specific sentences.
- The learner fixes it and sees the result change.

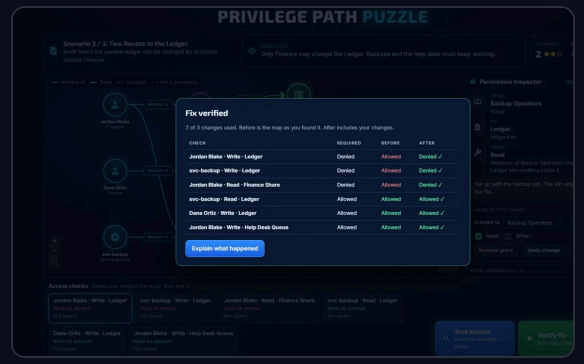
## Also in this folder

Download the reel or the slides as a PDF. Every game slide has a Play button.

# Twenty-two interfaces, one visual language. Dense when it needs to be, never confusing.



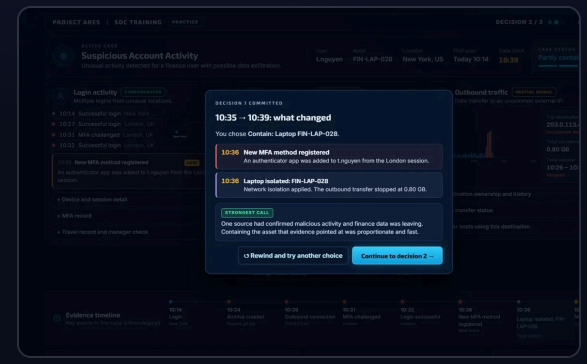

**Rule editor** with a live preview of who the rule allows and denies



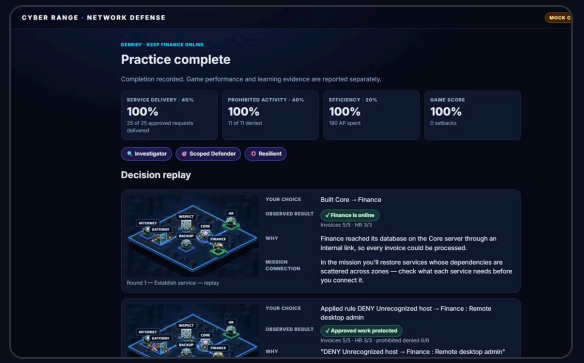

**Before and after table** that proves a fix kept business access



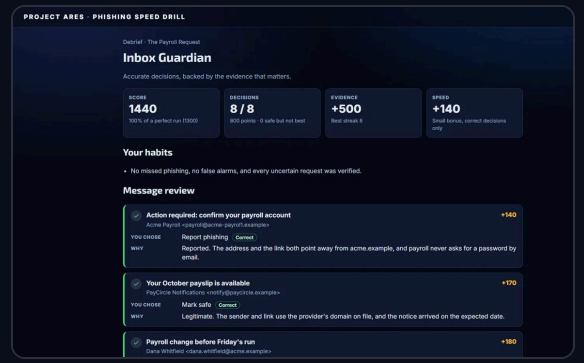

**Flight between stations**, collecting evidence on the way




**"What changed"** coach note, with rewind to any decision




**Decision replay** rebuilt from the learner's own moves



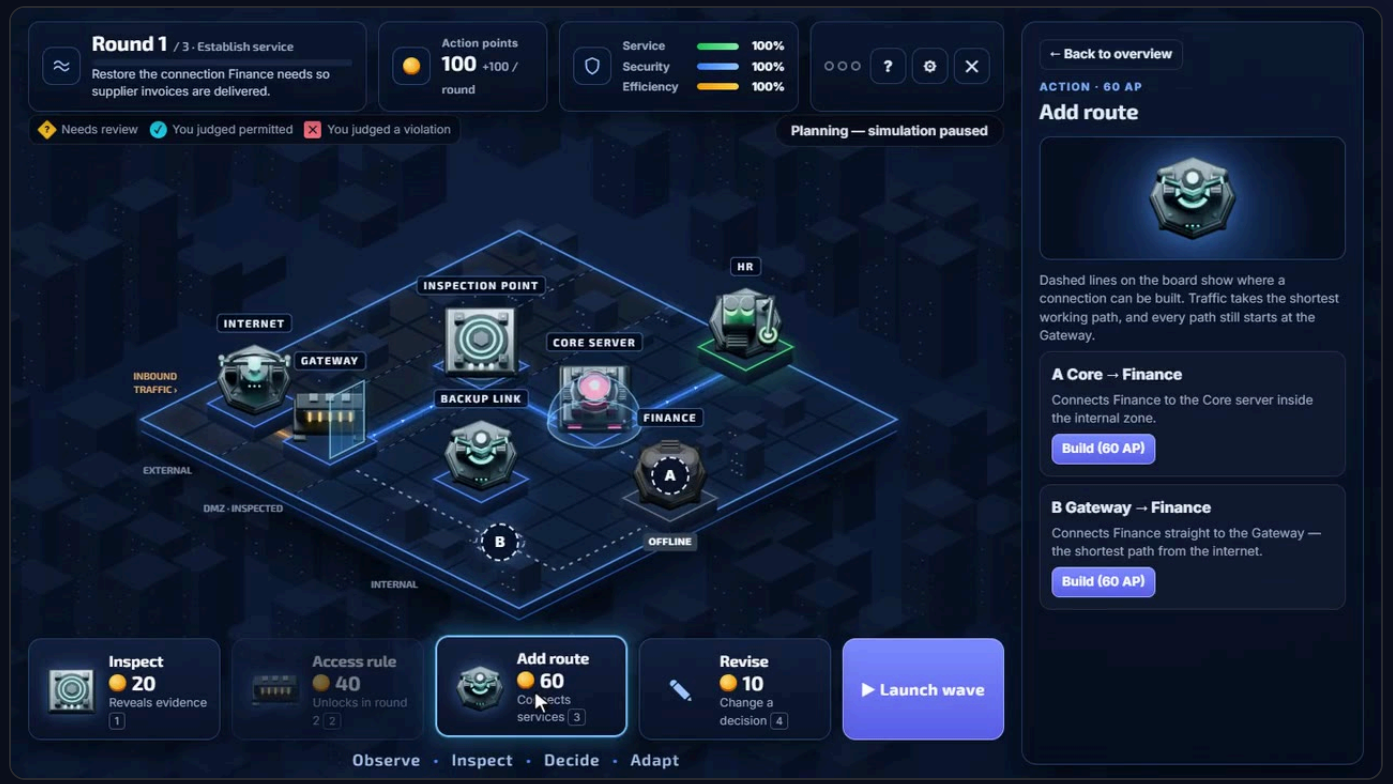

**Habits, not just a score:** false alarms, misses, skipped checks

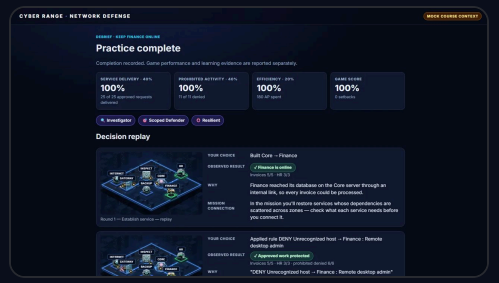



**Four-part score** with response time kept separate




**Feedback built from the simulation's own numbers**




**How it is scored.** 40% service delivered, 40% prohibited activity denied, 20% efficiency. Two reasoning checkpoints and the final challenge are recorded separately as learning evidence.

**Proved by tests.** “Blanket blocking cannot earn the best result” and “backup routes preserve the access policy”, each checked on 8 seeds. 32 tests.

**Depth and replay.** A new seed changes the traffic order, the attacker’s address and service, and which link fails.

GAME 1 OF 22 · ISOMETRIC STRATEGY

# Network Defense: Keep Finance Online

**Teaches:** How routes and dependencies keep a service up, how to read evidence, and how to write an access rule that stops the attacker without stopping the business.

EVIDENCE ON SCREEN

“Source 198.51.100.23 is not on any approved list and is outside the company network.” “Destination: Finance — Remote desktop admin (RDP 3389).” “Pattern: repeated sign-in attempts, each with a different username.”

AFTER A COMMON MISTAKE

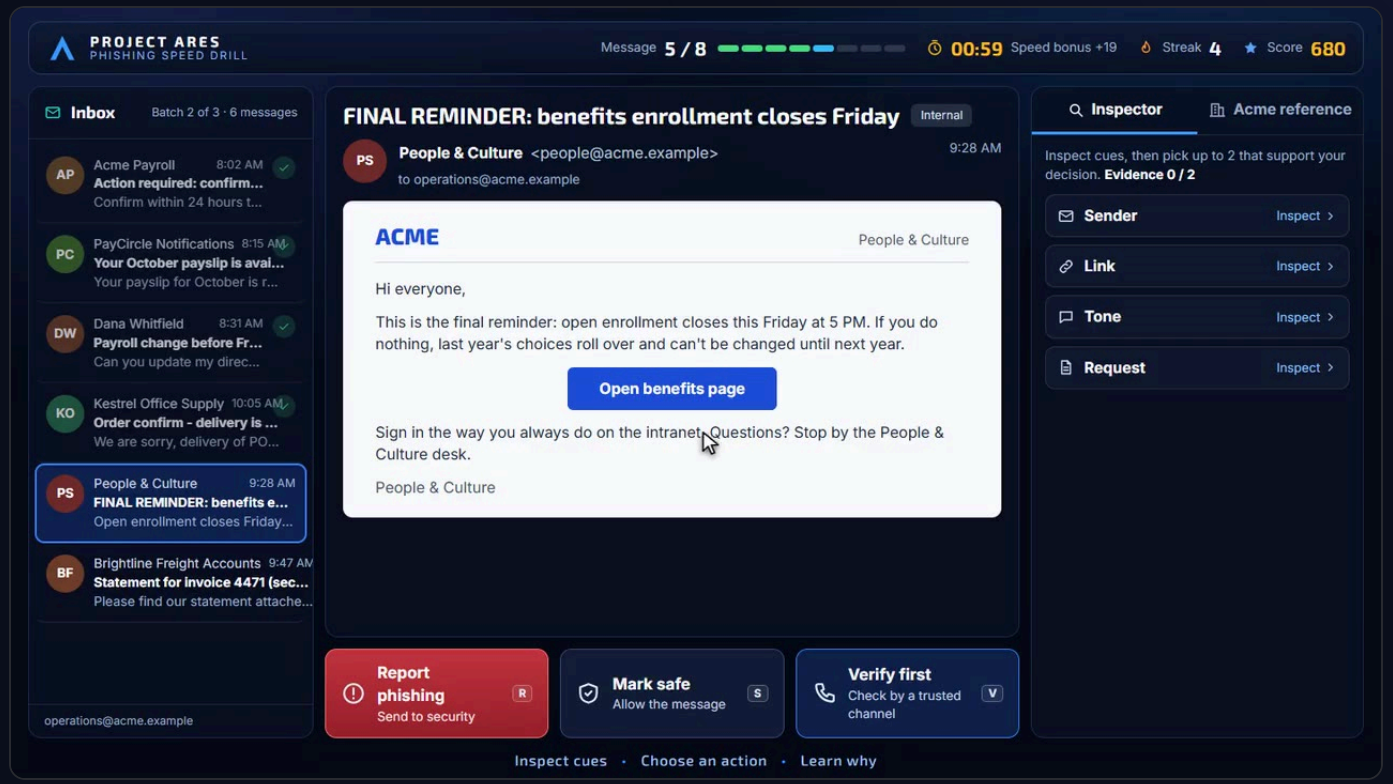
“You connected Finance straight to the Gateway. Invoices reached Finance, but Finance couldn’t reach its database on the Core server, so each one failed. Services depend on other services — a path from the internet isn’t enough.”

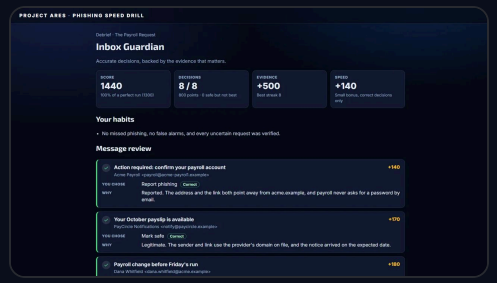
FINAL CHALLENGE, NO HINTS

“Which single rule best fits the policy?” Answer: `DENY 203.0.113.200 → HR : Admin console`

**Why it sticks.** The learner watches a rule succeed or fail on live traffic, so cause and effect are remembered together.

**Where it fits Ares.** Fits the reserved packet-gate and network-segmentation slots. Firewall and ACL rules are listed as untapped.




**How it is scored.** 100 for the best response, +20 per strong cue cited, a streak bonus, and a speed bonus that only counts after a correct decision. A false alarm scores zero.

**Proved by tests.** Always-report, always-safe and always-verify each "stay out of the top two ratings". "Slow but correct play still earns the top rating." 53 tests.

**Depth and replay.** 8 messages, 32 cues. The first message is guided and untimed; later batches are shuffled per seed.

GAME 2 OF 22 · TIMED INBOX DRILL

# Phishing Speed Drill: The Payroll Request

**Teaches:** Reading the sender, the reply-to and the real link destination, backing a call with evidence, and knowing when only a phone check can settle it.

> A LOOK-ALIKE DOMAIN
>
> `helpdesk@acme.example.it-notices.example` "A domain is read from the right. This address belongs to it-notices.example; 'acme.example' is only a label in front of it."

> A GENUINE ADDRESS, STILL NOT SAFE
>
> A bank-change request from a real colleague's address: "The address is genuine, but a matching address doesn't prove who is typing. Mailboxes get compromised."

> AFTER A FALSE ALARM
>
> "This was a real payslip notice. External mail and a generic greeting are not evidence by themselves; the domain, link, and timing all matched. False alarms slow down real work."

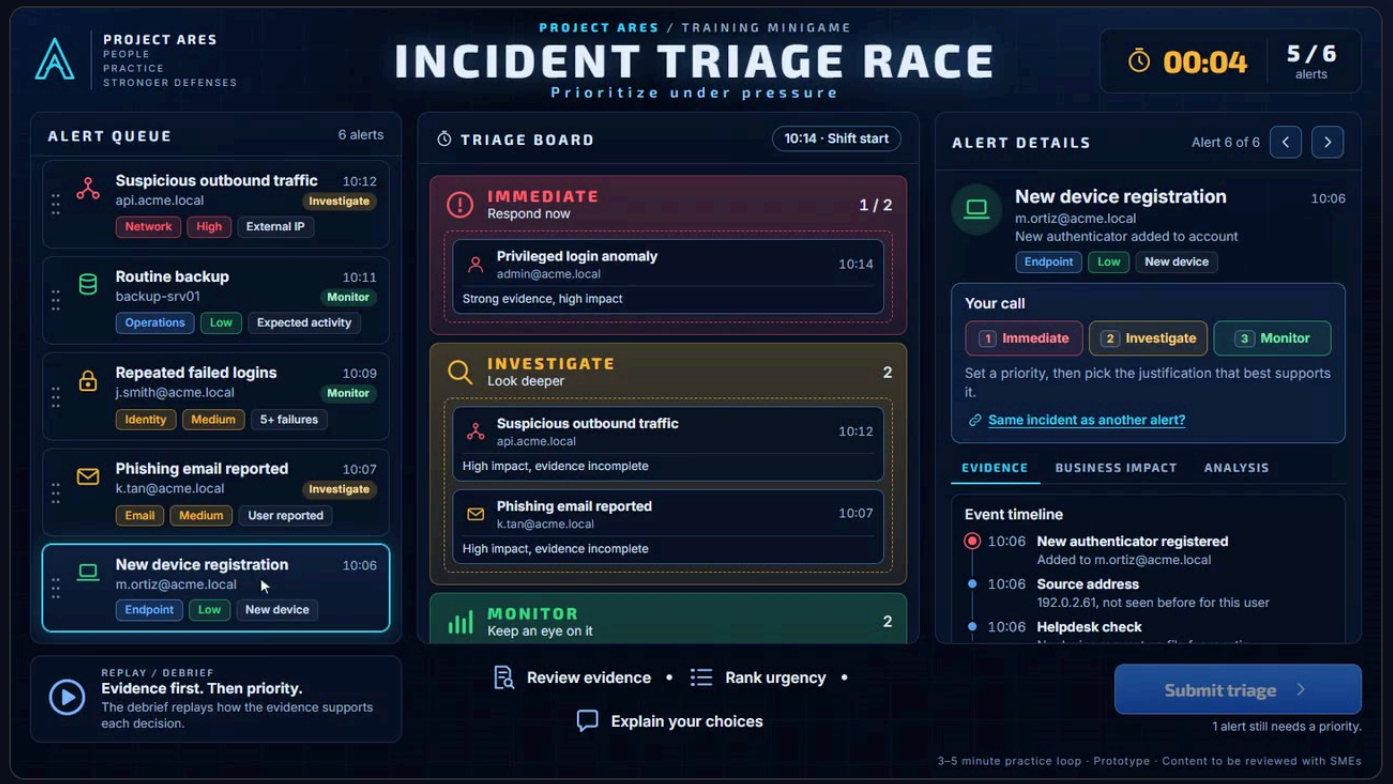
**Why it sticks.** Citing the evidence before deciding builds the habit of looking, not guessing.

**Where it fits Ares.** Fits the reserved phish-sprint slot. Suits awareness training for every employee, not only analysts.

GAMEPLAY RECORDED FROM THE WORKING BUILD · QUOTES FROM THE GAME'S DATA FILE

# Incident Triage Race: The Busy Morning

**Teaches:** Setting priority from evidence and business impact, linking alerts that are one incident, and revising when new facts arrive.




### TAGGED "LOW" IN THE QUEUE

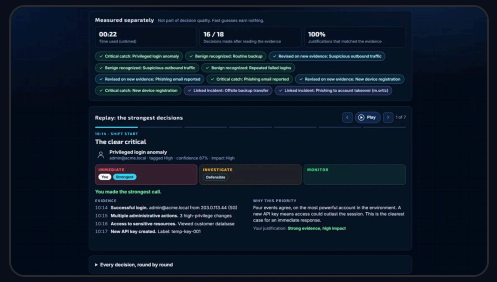
A new device on an account that "approves outgoing payments". "The quietest alert in the queue turns out to be an attacker with a working sign-in on the payments portal. Business impact, not the severity tag, sets the urgency."

### A MID-SHIFT UPDATE

"Infrastructure confirms change ticket CHG-4471: backups moved to a new offsite storage provider this morning… Nothing else in the queue is explained by it."

### A JUSTIFICATION THAT ALWAYS SCORES ZERO

"The severity tag says so."



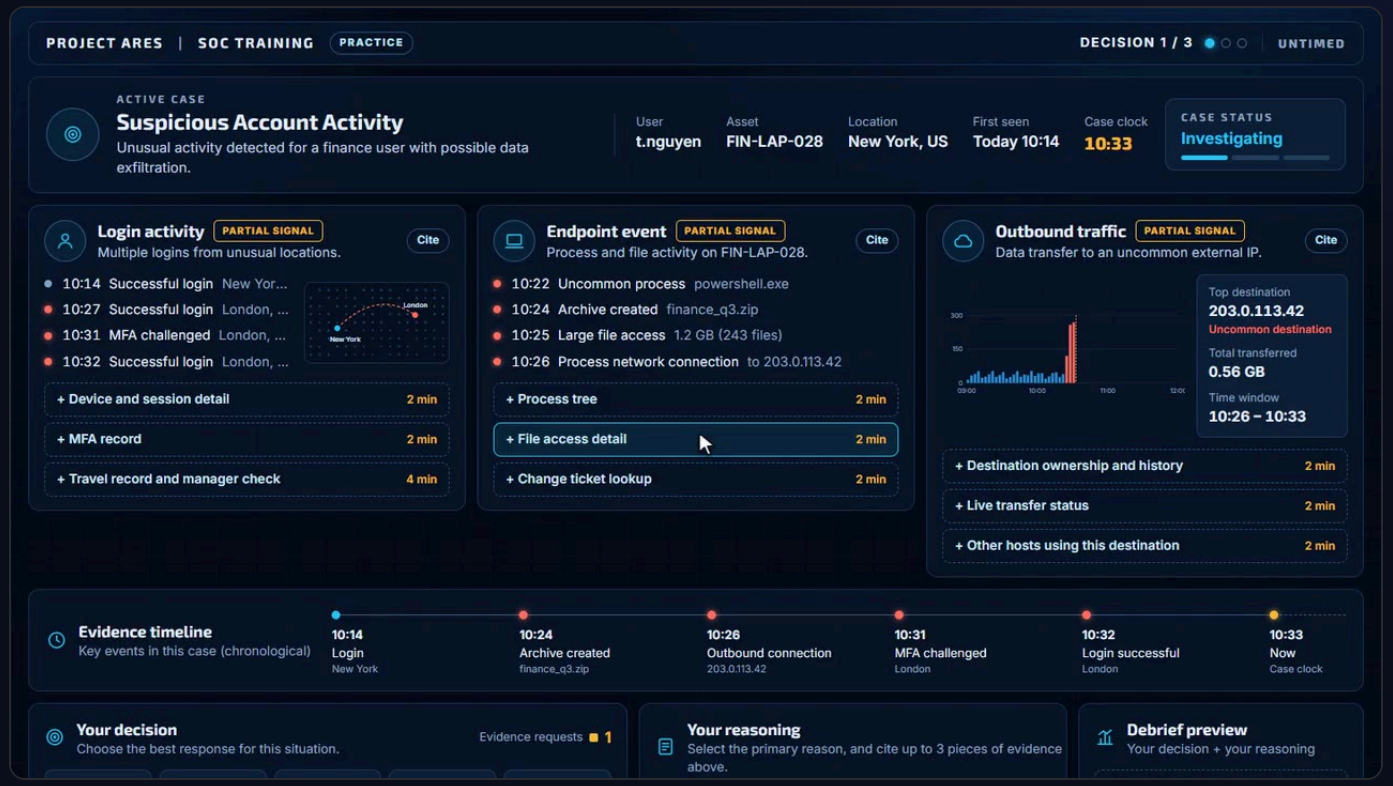
**How it is scored.** Points for the best or a defensible lane, a matching reason, revising on new evidence and correct links. Missed criticals and needless escalations cost points. Time is reported separately.

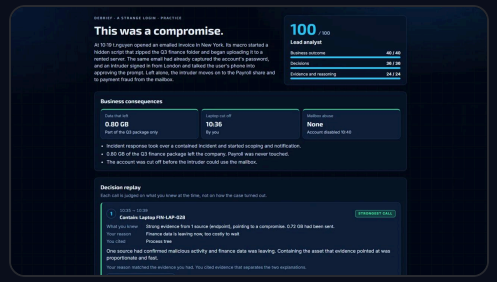
**Proved by tests.** "Never treats the severity tag as a valid justification." "Rewards revising when evidence changes, not standing still." 13 tests.

**Depth and replay.** 6 alerts, 29 evidence items, 2 updates over 3 rounds, with a 7-moment replay in the debrief.

**Why it sticks.** Being wrong about the quiet alert, then seeing why, is the lesson people repeat to colleagues.

**Where it fits Ares.** Matches the design guide's "Alert Triage" concept. A natural team game for workshops.




**How it is scored.** Outcome 40, decisions 36, reasoning 24. “Each call is judged on what you knew at the time, not on how the case turned out.”

**Proved by tests.** “No single blind opening wins in both variants.” “Investigating first is not always right: delay lets the incident progress.” 13 tests.

**Depth and replay.** The seed decides whether it is a compromise or approved work. Practice mode can rewind to any decision.

GAME 4 OF 22 · SOC INVESTIGATION CASE

# SOC Dashboard Challenge: A Strange Login

**Teaches:** Separating evidence that discriminates between explanations from evidence that fits both, and responding in proportion while data is leaving.

### SAME OPENING, TWO POSSIBLE TRUTHS

Compromise: “OUTLOOK.EXE → WINWORD.EXE (Invoice_Q3_revised.docm) → powershell.exe -WindowStyle Hidden -EncodedCommand …” Approved work: “Task Scheduler → powershell.exe -File QuarterClose-Export.ps1”.

### EVIDENCE THAT PROVES NOTHING

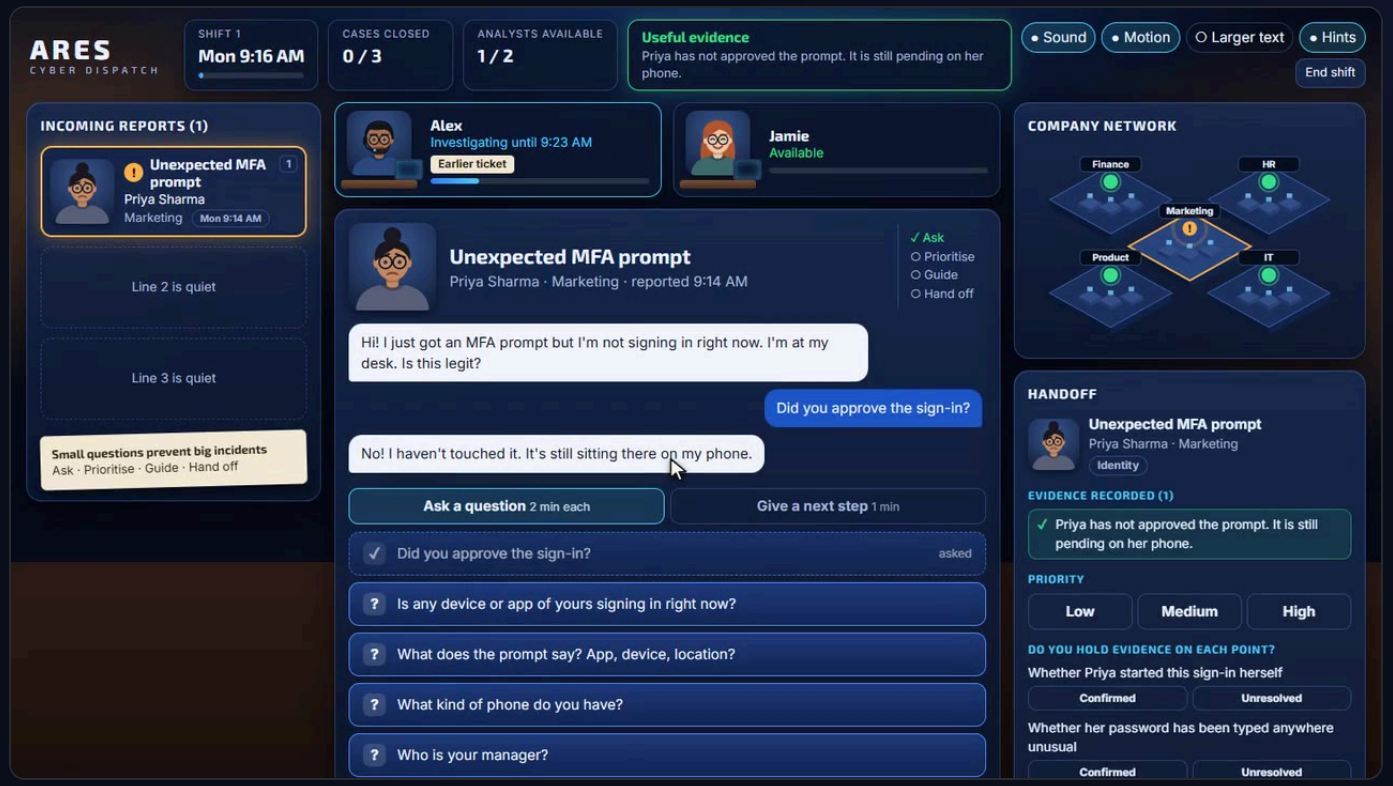
“A passed prompt shows that someone tapped Approve. Intruders get prompts approved too, so this fits both explanations.”

### AFTER OVER-REACTING

“You had strong evidence that this was approved work and contained anyway. The business paid for an interruption the evidence did not support.”

**Why it sticks.** Two truths with the same opening teach that the right call comes from evidence, not from matching a familiar story.

**Where it fits Ares.** A step up from triage toward analyst missions, and a warm-up before a range investigation.







**How it is scored.** Four parts of 25 per report: evidence gathered, priority, guidance and handoff. Response time is shown but kept out of the score.

**Proved by tests.** "Escalating everything as high scores poorly." "Asking every question before acting is not the best run." 24 tests.

**Depth and replay.** 6 reports over 2 shifts, 38 questions, 24 guidance options. The second shift changes which report is benign.

GAME 5 OF 22 · DISPATCHER DESK

# Cyber Dispatch: First Shift

**Teaches:** Asking the questions that matter, giving the safe next step before harm lands, and handing off only what the evidence confirms.

### A REPORT COMES IN

"Hi! I just got an MFA prompt but I'm not signing in right now. I'm at my desk. Is this legit?" Asking reveals: "The prompt shows a Mail sign-in from Rotterdam. She is in the Lisbon office."

### THE SAFE STEP, IN TIME

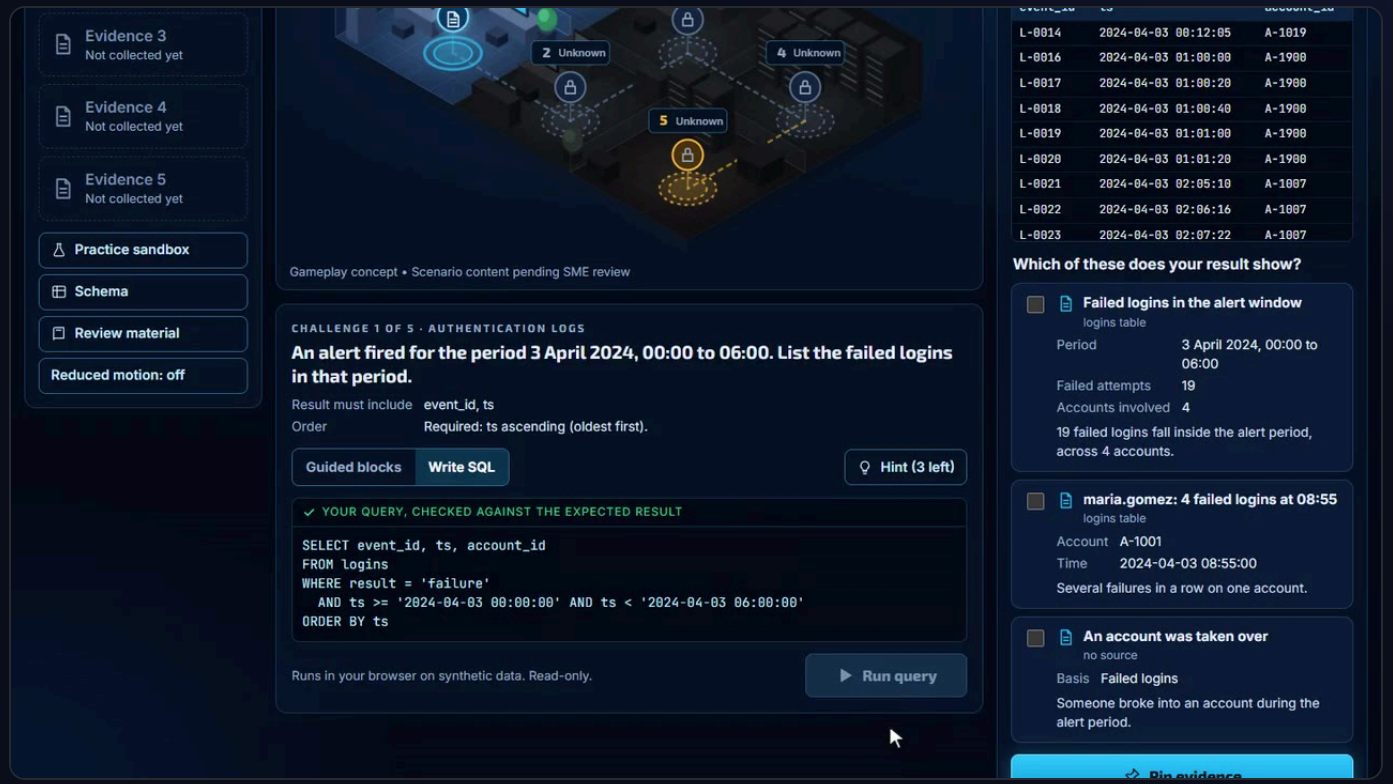
"Turn off Wi-Fi and unplug the network cable now. Leave the laptop switched on."

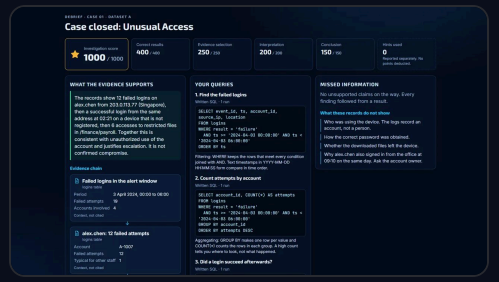
### A HARMFUL STEP AND ITS CONSEQUENCE

"Reply to the email and ask them to confirm the new details." → "The sender 'confirmed' their own request, and Elena queued the payment."

**Why it sticks.** Recoverable harm: the learner sees the damage a bad instruction causes, then gets to put it right.

**Where it fits Ares.** Covers first response and handoff, part of the incident response lifecycle listed as untapped. Good for help desks.




**How it is scored.** Out of 1,000: correct results, evidence cards, interpretation and conclusion. A right conclusion without its evidence scores zero. Hints are counted, never deducted.

**Proved by tests.** “Accepts equivalent queries written differently.” “Never grants evidence for running a query.” Real SQLite runs in the tests. 31 tests.

**Depth and replay.** A second dataset makes the highest failure count the wrong account to follow. Guided tiles or written SQL.

GAME 6 OF 22 · SQL INVESTIGATION

# Query Hunt: Unusual Access

**Teaches:** Filtering, aggregating and joining real log tables, reading the result correctly, and telling a suspicious observation from a confirmed conclusion.

### A QUERY THE LEARNER PRODUCES

```
SELECT account_id, COUNT(*) AS attempts FROM logins WHERE result = 'failure' … GROUP BY account_id ORDER BY attempts DESC
```

### WHEN THE LEARNER JUMPS AHEAD

“Repeated failed logins alone do not establish compromise: they show attempts. Which evidence shows what happened after them?”

### THE DEBRIEF NAMES THE LIMITS

“The records do not show who was at the keyboard, how the password was obtained, or that data left the device.”

**Why it sticks.** Writing the query is retrieval practice. Being stopped from over-claiming teaches analytic discipline.

**Where it fits Ares.** Fits the reserved log-hunt slot. Log analysis is listed as untapped.







**How it is scored.** Continuity 35%, policy 30%, resources 15%, justification 20%. A failed access review marks the security objective failed whatever the score.

**Proved by tests.** “Adding capacity helps only where capacity is the constraint.” “Does not let a high score hide a failed security objective.” 18 tests.

**Depth and replay.** 3 checkpoints, 11 evidence items. Fly manually or on autopilot; flying is reported apart from the learning score.

GAME 7 OF 22 · ARCADE FLIGHT

# Cloud Command: First Orbit

**Teaches:** Finding the real capacity constraint through a dependency, balancing availability against cost, least privilege, and who is responsible for what on IaaS.

### REQUIREMENTS

“Serve at least 99% of checkout requests at the peak.” “Keep 10% headroom above the expected peak on every service in the request path.” “Cost ceiling: 120 credits for the 2-hour surge window.”

### AFTER SCALING THE WRONG SERVICE

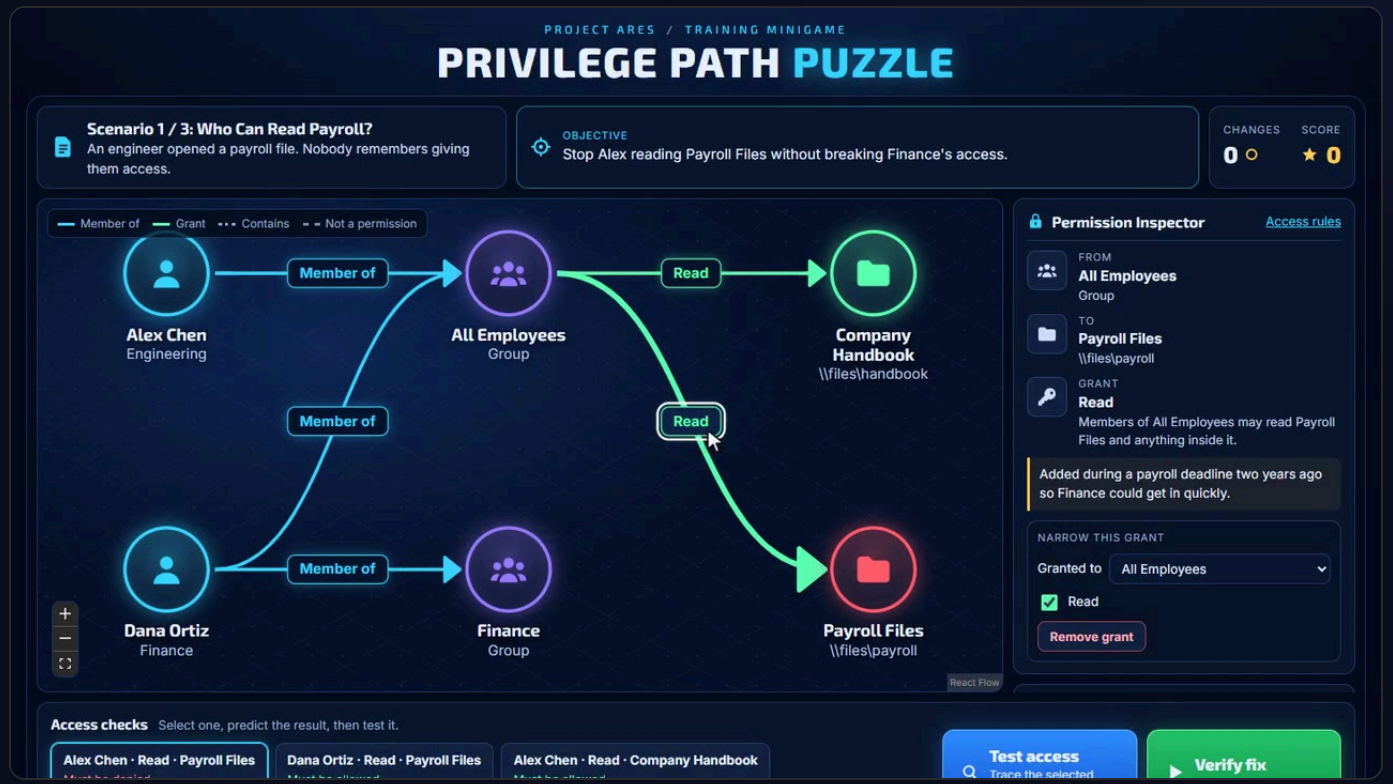
Checkout “has enough capacity, but each request calls” the Payments API, “which can serve only” 1,200 rps. More checkout instances “cannot help until” it has capacity.

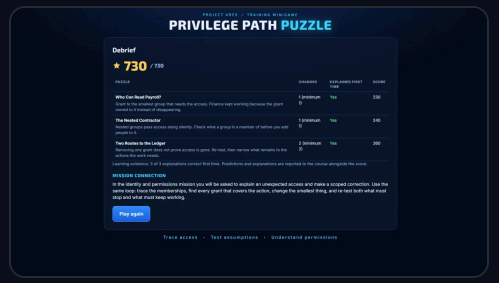
### LEAST PRIVILEGE

A backup account holds 7 permissions. “Reading a snapshot requires keys:use on that key. The ticket does not mention it.”

**Why it sticks.** The numbers in the feedback come from the simulation, so the explanation matches exactly what the learner did.

**Where it fits Ares.** Cloud security is listed as untapped. The arcade layer shows the same content can carry a lighter style.




**How it is scored.** 100 for solving, a bonus for matching the minimum number of changes, points for correct predictions, and 40 for explaining it right first time.

**Proved by tests.** “Never treats connectivity as authorization.” Each puzzle “cannot be solved with fewer changes than par” and “fails when access is removed for everyone”. 25 tests.

**Depth and replay.** 3 puzzles, 22 nodes, 13 access checks. Inspect, predict, test, change, verify, explain.

GAME 8 OF 22 · ACCESS GRAPH PUZZLE

# Privilege Path Puzzle: Who Can Read Payroll

**Teaches:** Tracing nested groups and inherited grants to find every route to a file, then making the smallest change that closes it without breaking business access.

### THE BRIEF

“An engineer opened a payroll file. Nobody remembers giving them access.” Hidden in the graph: “Added during a payroll deadline two years ago so Finance could get in quickly.”

### A CHECK THE FIX MUST STILL PASS

“svc-backup · Write · Ledger — must be denied”: “A backup job copies data. It never needs to change it.”

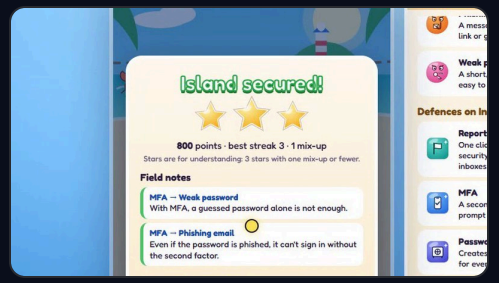
### AFTER A WRONG EXPLANATION

“Signing in is authentication: it proves who Alex is. What Alex may open is authorization, and that came from a grant.”

**Why it sticks.** Predict, test, then explain: the learner commits to an answer before seeing the result.

**Where it fits Ares.** Matches the design guide’s “IAM Auditor” concept. Identity and access is listed as untapped.




**How it is scored.** Points per match with a streak bonus. Stars count mix-ups, not speed: three stars means one mix-up or fewer.

**Proved by tests.** "Starts every level solvable: each threat has a matching defence in the pile." The game drops in a missing defence so no level can get stuck. 10 tests.

**Depth and replay.** 5 islands, a tide of new threats from island 2, iced "unidentified" alerts from island 4, and a field guide whose match book fills in as pairs are found.

GAME 9 OF 22 · PHYSICS STICKER PILE

# Secure Stack: Five Islands

**Teaches:** Which control actually stops which threat, that one control can cover several threats, and investigating an unknown alert before acting on it.

A RIGHT MATCH

"Report button + Phishing email. Reported fast, the email is pulled from every inbox before more people click."

AN UNIDENTIFIED ALERT

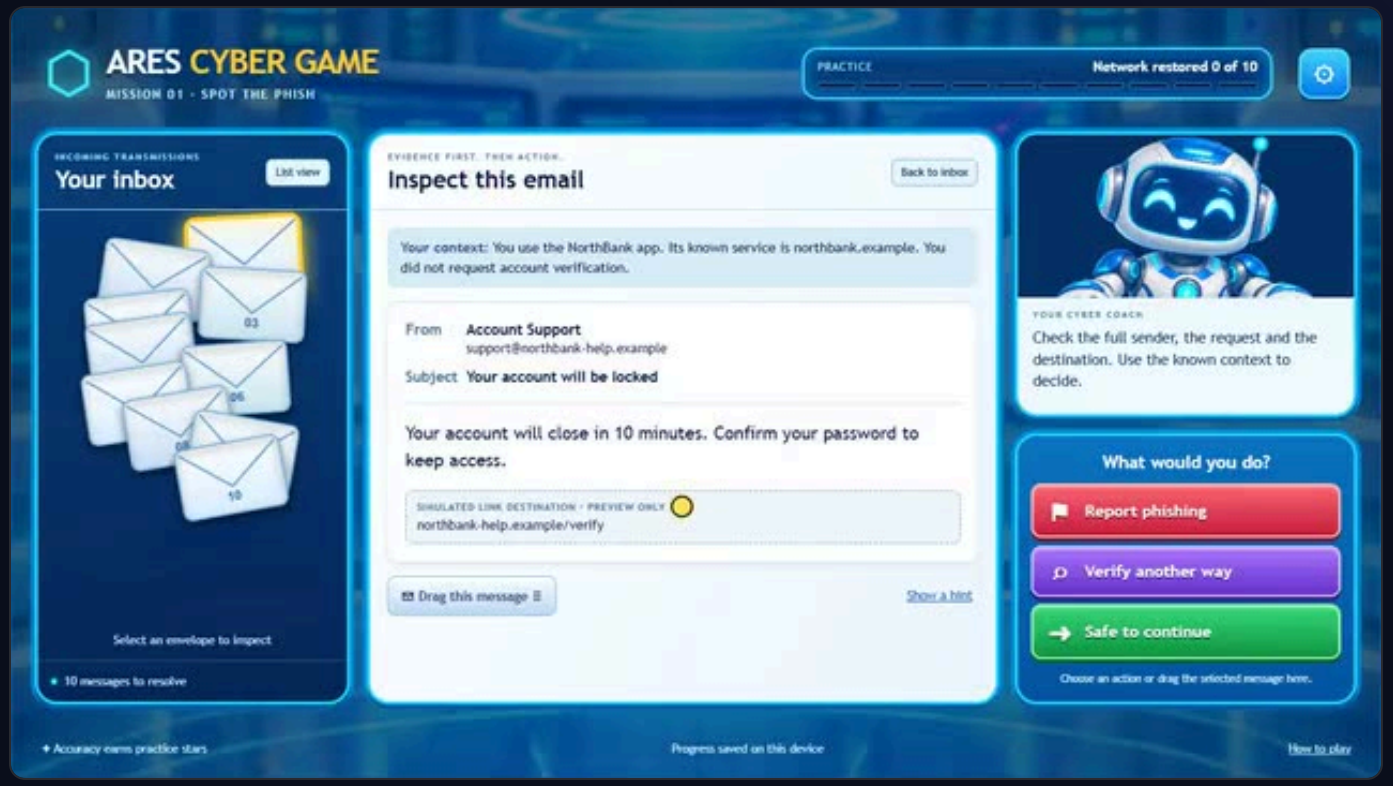
"Nobody knows what this alert is yet. Secure a threat right next to it to investigate it."

AFTER A WRONG DROP

"Password manager stops weak password, not phishing email. Phishing email needs Report button or MFA."

**Why it sticks.** The match rule is the lesson: every drop is a claim about which control works, and the crab answers it at once.

**Where it fits Ares.** All-staff awareness that plays on a phone. A two-minute warm-up before a mission.







**How it is scored.** Accuracy earns practice stars. Each right call restores a node of the network, ten in all, then four transfer messages test the habit.

**Proved by tests.** Built and tested separately, and hosted on its own site; it is the one game here that is not in this repository.

**Depth and replay.** 10 messages with a robot coach and hints, drag-to-decide or buttons, and progress saved on the device.

GAME 10 OF 22 · INBOX WITH A COACH

# Spot the Phish: Mission 01

**Teaches:** Reading a message against what you already know, and choosing between reporting, verifying another way and continuing safely.

YOUR CONTEXT

"Your IT team never asks for your authentication codes. You did not request support."

THE REQUEST

"Reply with the six-digit code from your authenticator to prevent account suspension."

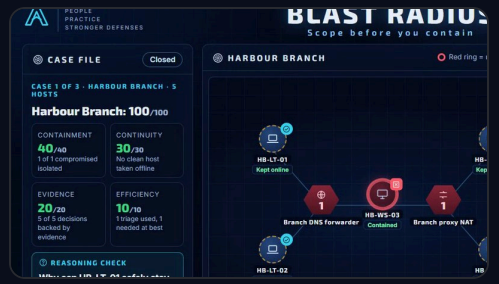
AFTER A WRONG CALL

"Your first attempt is recorded. You can try again without losing a life or waiting."

**Why it sticks.** Every message is read against a context line, so the learner judges the request, not the logo.

**Where it fits Ares.** Fills the same gap as the reserved phish-sprint slot, for every employee.




**How it is scored.** Containment 40, continuity 30 (a clean critical server taken offline counts three times), evidence support 20, triage efficiency 10. A right guess scores 70.

**Proved by tests.** “Never settles a case from the sensors alone, but always within the triage budget.” “Keeps blanket strategies well under 60.” 20 tests.

**Depth and replay.** 3 cases with two truths each; Retry swaps the truth. What the evidence proves is checked against every possible arrangement of hosts.

GAME 11 OF 22 · DEDUCTION PUZZLE

# Blast Radius: The Backup Account

**Teaches:** Scoping a compromise from sensors that only count, spending expensive triage where it settles the most, and saying “unknown” honestly.

### WHY A SENSOR ONLY COUNTS

“Branch clients resolve through this forwarder, so the central DNS log shows only the forwarder’s address.”

### THE DEDUCTION, IN THE DEBRIEF

“Branch DNS forwarder counted 1, and HB-WS-03 already accounts for it.”

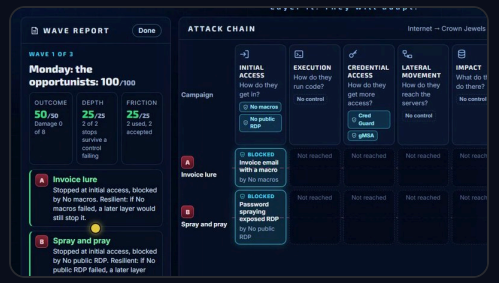
### AFTER A LUCKY GUESS

“Contained, but it was a guess. The evidence you gathered left it open; a triage or another sensor would have settled it.”

**Why it sticks.** A guess that happens to be right scores less than a deduction, so learners learn to prove before they isolate.

**Where it fits Ares.** Scoping, the step after SOC Dashboard in the incident response lifecycle.




**How it is scored.** Outcome 50 (damage, footholds included), depth 25 (stops that survive any one control failing), friction 25 (within what the business accepts).

**Proved by tests.** "Detection without response leaves the damage in place." "Perimeter-only planning collapses when the actors adapt." 21 tests.

**Depth and replay.** 3 waves, 16 ATT&CK techniques, 15 controls. Attackers switch to fallbacks; Retry changes their preferences.

GAME 12 OF 22 · DEFENCE PLANNING BOARD

# Defense in Depth: The Long Week

**Teaches:** Mapping controls to the attack stages they affect, layering so no single failure lets an attack through, and knowing that detection needs response.

### A CONTROL CARD

Phishing-resistant MFA: "The key checks the site it is talking to, so a relay page gets nothing it can replay."

### THE REASONING CHECK

"If Block macros in files from the internet failed, what would stop 'Invoice lure'?" The game re-runs the attack to answer.

### A SINGLE POINT OF FAILURE

"Stopped at initial access, blocked by No macros. Single point of failure: without No macros it gets through."

**Why it sticks.** The question every defender should ask, "what if this control failed?", is asked after every wave and answered by the simulation.

**Where it fits Ares.** Security architecture and ATT&CK-informed defence. Suits managers who approve security budgets.







**How it is scored.** Named it 30, response 40 (graded against the best card still in your toolkit), outcome 30.

**Proved by tests.** “Separates naming the technique from choosing the response.” “Keeps naive strategies well below the best play.” 15 tests.

**Depth and replay.** 3 incidents, 16 techniques, 22 response cards. The actor switches technique when countered; Retry changes hosts, users and addresses.

GAME 13 OF 22 · CARD DUEL

# Counterplay: Three Incidents

**Teaches:** Reading raw telemetry, telling a technique from its look-alikes, and choosing the response that removes the attacker’s capability.

> TELEMETRY, FACE DOWN
>
> `idp-signin user=j.alvarez result=success mfa=satisfied by claim in token device=unregistered`

> THE TELL, AT THE REVEAL
>
> “A successful sign-in with MFA ‘satisfied by claim in token’ from an unregistered device: a stolen session.”

> A PARTIAL FIX
>
> “Block the sender domain: Stops new messages, but the ones already delivered stay in inboxes and keep getting clicked.”

**Why it sticks.** Naming the technique before responding makes learners read the log line, not reach for the scariest-sounding card.

**Where it fits Ares.** The incident response lifecycle, listed as untapped. A step into SOC tier 1 work.







**How it is scored.** 10 points per task right first time, 6 on the second check, 3 after that, 0 if the answer is shown. Hints are free.

**Proved by tests.** “Sizes subnets to the smallest that fits.” “Explains each kind of wrong answer specifically.” 8 tests.

**Depth and replay.** 10 tasks across binary, masks, sizing and finding the network with AND.

GAME 14 OF 22 · GEAR WORKBENCH

# Bit by Bit: The New Depot

**Teaches:** Binary octets, masks from prefixes, the smallest subnet that fits, and the network address of a host named in an alert.

THE TASK

“An alert names 10.20.4.77/26. Build the network address it belongs to.”

RIGHT ANSWER

“77 AND 192 = 64. The network is .64/26; hosts run from .65 to .126.”

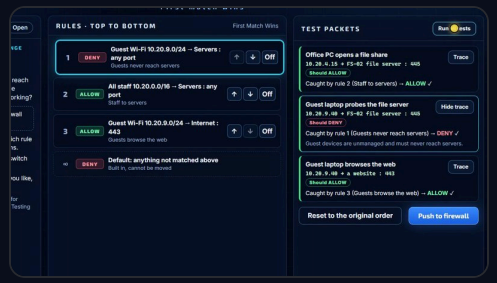
TOO BIG A SUBNET

“/25 fits (126 usable) but wastes addresses. A smaller subnet still fits 50: add a network bit.”

**Why it sticks.** The answer is built bit by bit on gears, so the maths becomes something you can see and touch.

**Where it fits Ares.** Networking fundamentals, ahead of Network Defense, Rule Order and Blast Radius.



**How it is scored.** Correct 70 (packets with the right verdict), tidy 30 (fewest changes, measured as edit distance). Tracing and testing are free.

**Proved by tests.** “Does not let the cleanup rule simply be deleted on an allow-by-default firewall.” “Does not solve the legacy rule by reordering alone.” 9 tests.

**Depth and replay.** 4 change tickets with test packets. The fewest changes for each ticket (1, 2, 1 and 3) are found by search.

GAME 15 OF 22 · RULE-ORDERING PUZZLE

# Rule Order: The FW-CORE Change Window

**Teaches:** First-match firewall logic, shadowed rules, where the cleanup rule belongs and what the default does, and removing over-broad rules.

> TRACING A PACKET
> “Caught by rule 1 (Staff to servers) → ALLOW”, for a guest laptop that should have been denied.

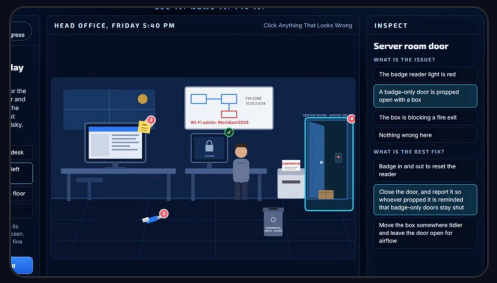
> THE LESSON
> “A broad allow above a narrow deny hides the deny completely: this is called shadowing.”

> THE TEMPTING DELETE
> “A cleanup ‘deny everything’ rule belongs at the very bottom… And on a firewall whose default is allow, deleting it opens everything up.”

**Why it sticks.** Tracing a packet makes the firewall’s reading order visible, so “first match wins” is seen, not memorised.

**Where it fits Ares.** Firewall and ACL rules, listed as untapped; the reserved packet-gate slot.




**How it is scored.** 4 points for finding each risk, 3 for naming the issue, 3 for the right fix. Each false alarm costs 5.

**Proved by tests.** “Charges false alarms, so flagging everything does not win.” “Credits finding a risk even with the wrong reason, but less.” 6 tests.

**Depth and replay.** 2 illustrated scenes, 13 risks and 4 decoys, each with an issue, a fix and an explanation.

GAME 16 OF 22 · HIDDEN-OBJECT SCENE

# Spot the Risks: The Walk-round

**Teaches:** Physical and social-engineering risks at the office and at home, naming the real issue, choosing the fix over a workaround, and leaving good practice alone.

### A DROPPED USB STICK

“Dropped USB drives with tempting labels are a long-standing attack. Plugging one in, even ‘just to check’, can run code.”

### A DECOY

“A locked screen is exactly what good looks like. Flagging it costs time and trust.”

### A TEMPTING WRONG FIX

“Move the note under the keyboard.” The note reads a VPN password: “Anyone walking the floor, including cleaners and visitors, now has a working password.”

**Why it sticks.** The decoys teach what good looks like, so the habit is judgement rather than suspicion of everything.

**Where it fits Ares.** All-staff awareness and onboarding. Not on the design guide’s list yet.







**How it is scored.** 10 points per detection, minus 3 per hint. Showing a reference pattern scores 0 but teaches the trick.

**Proved by tests.** “Every reference pattern solves its level.” “The unescaped address falls for the look-alike.” 5 tests.

**Depth and replay.** 8 levels, each built around one regex idea and one trap. Lines light up as you type: caught, missed, false alarm.

GAME 17 OF 22 · REGEX WORKBENCH · REMIX OF REGEXILE

# Pattern Patrol: Catch the Attack, Not the Noise

**Teaches:** Writing a detection that catches every attack line and none of its benign look-alikes: escaping, classes, alternation, quantifiers, boundaries.

> A FALSE ALARM, ON THE LINE IT HITS
>
> `Failed password for administrator-training`: “Contains ‘admin’ but is a training account.”

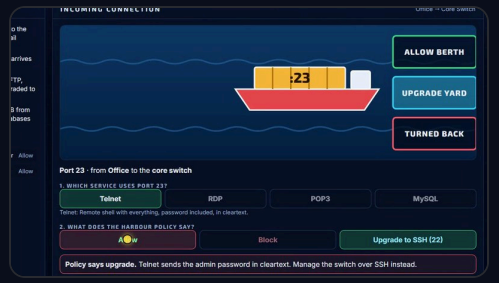
> THE FIX
>
> `Failed password for (root|admin)\b`: the word boundary removes the noise.

> THE LESSON ON DOTS
>
> “In a regex a dot means ‘any character’. Escape it as \. when you mean a real dot, and use \b so .44 doesn’t also match .440.”

**Why it sticks.** Every false alarm shows on the exact line it hits, so precision stops being abstract.

**Where it fits Ares.** Log analysis and SIEM, listed as untapped. Credited remix of RegExile by the games team.




**How it is scored.** 1 point for the service, 3 for the policy decision. Knowing every port but allowing everything stays under 60.

**Proved by tests.** “Only asks for an upgrade where a secure version exists.” “Covers all three decisions and scores 100 for a perfect run.” 3 tests.

**Depth and replay.** 15 ships over three tides, covering Telnet → SSH, FTP → SFTP, HTTP → HTTPS, POP3 → POP3S and LDAP → LDAPS.

GAME 18 OF 22 · HARBOUR SORTING · REMIX OF PORTFLOW

# Port Authority: Every Port Tells You Who Is Knocking

**Teaches:** Naming the service from its port, then applying a policy: allow it, block it, or upgrade cleartext to its secure version.

THE HARBOUR RULE

“Admin access (SSH, RDP) only arrives through the VPN.”

SAME PORT, RIGHT SOURCE

“RDP is acceptable when it arrives through the VPN to the jump host.”

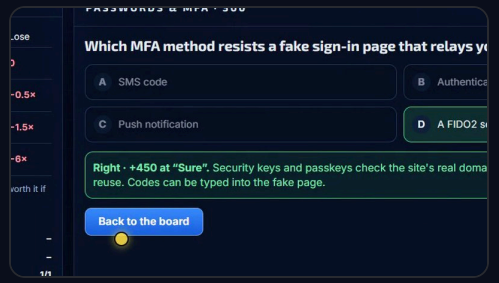
AFTER ALLOWING TELNET

“Policy says upgrade. Telnet sends the admin password in cleartext. Manage the switch over SSH instead.”

**Why it sticks.** The same port is right from one place and wrong from another, so learners read the source, not just the number.

**Where it fits Ares.** Firewall rules and the packet-gate slot. Credited remix of Portflow by the games team.




**How it is scored.** Unsure +½ / 0, Likely +1 / −½, Sure +1½ / −1½, Certain +2 / −6, times the tile's value.

**Proved by tests.** "Makes each confidence level the best bet in its own accuracy band." 3 tests.

**Depth and replay.** 24 questions in 6 categories, each with an explanation, and a calibration chart in the debrief.

GAME 19 OF 22 · QUIZ BOARD · REMIX OF TRIVIALOOT

# Confidence Board: Know What You Know

**Teaches:** Six areas of security knowledge, and calibration: how sure you should be before you act on what you think.

> THE WAGER
> "'Certain' wins 2× but loses 6×: only worth it if you'd be right 9 times in 10."

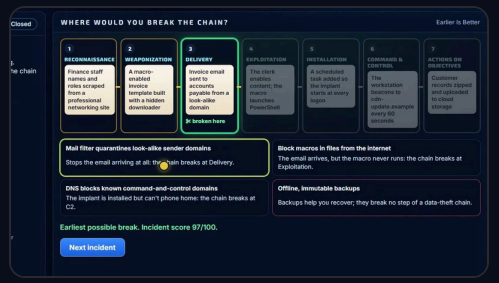
> A QUESTION
> "Which MFA method resists a fake sign-in page that relays your codes?" A FIDO2 security key or passkey.

> THE DEBRIEF
> "Overconfident at 'Certain'… Save 'Certain' for things you could explain to someone else."

**Why it sticks.** An analyst who says "certain" must be right, and saying "not sure, let me verify" becomes a skill, not a weakness.

**Where it fits Ares.** A knowledge check across domains. Credited remix of Trivialoot by the games team.



**How it is scored.** Placement 70 (minus 3 per mistake), then 30 for the earliest break, 15 for a later valid one, 0 for one that breaks nothing.

**Proved by tests.** “Each incident has one card per stage and at least one break that does nothing.” “Scores the earliest break best and an irrelevant one at zero.” 3 tests.

**Depth and replay.** 3 incidents: an invoice lure, an exposed VPN and a poisoned update.

GAME 20 OF 22 · CARD PATIENCE · REMIX OF CYLITAIRE

# Chain Breaker: Rebuild It, Then Break It Early

**Teaches:** Placing real attack activity on the kill chain, seeing that later links depend on earlier ones, and telling a control that breaks the chain from one that only helps you recover.

A MISPLACED CARD

“Not Delivery: this step happens earlier in the chain.”

THE EARLIEST BREAK

“Mail filter quarantines look-alike sender domains: Stops the email arriving at all: the chain breaks at Delivery.”

A BREAK THAT BREAKS NOTHING

“Offline, immutable backups: Backups help you recover; they break no step of a data-theft chain.”

**Why it sticks.** Every link after the break visibly never happens, which makes “stop it early” concrete.

**Where it fits Ares.** The kill chain, alongside the hub’s game. Credited remix of Cylitaire: Kill Chain by the games team.






**How it is scored.** 100 for a first-try tag, 50 for a second, 0 after that or if the ship gets through.

**Proved by tests.** “Uses every tactic and only known techniques.” “Accepts any tactic a multi-tactic technique serves.” 3 tests.

**Depth and replay.** 18 techniques over 3 waves in rough intrusion order. Calm mode, on by default, makes it about knowledge, not reflexes.

GAME 21 OF 22 · ORBITAL DEFENCE · REMIX OF ASTRO-DEFENSE

# Tactic Orbit: Name the Tactic, Raise the Shield

**Teaches:** Reading an ATT&CK technique and naming the tactic it serves: the attacker’s “why” behind each action.

TARGET INTEL

T1053.005 Scheduled task: “A task is created to run the payload on a schedule.”

WHY SEVERAL SHIELDS WORK

“Scheduled tasks serve several tactics: they run code (Execution), survive reboots (Persistence), and can run as SYSTEM (Privilege Escalation).”

AFTER A WRONG SHIELD

“Not Exfiltration. The victim opens a file that runs the payload. Think about what this step achieves for the attacker.”

**Why it sticks.** Ships arrive roughly in the order of a real intrusion, so the tactics form a story, not a list.

**Where it fits Ares.** ATT&CK tactics, alongside the hub’s game. Credited remix of Astro-Defense by the games team.







**How it is scored.** Per attack path: up to 70 for how early it was first seen, plus 30 if two or more sensors saw it.

**Proved by tests.** “Can’t buy everything, and the best plan is a real trade-off.” “Rewards early detection over late.” 3 tests.

**Depth and replay.** 9 sensors, 3 attack paths, and the best plan for the budget found by checking all 512 combinations.

GAME 22 OF 22 · SENSOR PLANNING · REMIX OF THREAT MODELER

# Blind Spots: You Can’t Stop What You Can’t See

**Teaches:** Placing detection on attack paths within a budget, and why seeing an attack early and twice beats watching only the exits.

A SENSOR

Attachment sandbox, 2 points: “Detonates attachments and flags the ones that misbehave.”

A PATH, AFTER THE RUN

“First seen at step 1 of 7. Seen more than once.”

AFTER WATCHING ONLY THE EXITS

“Watching only the exits catches thieves on the way out, after the damage is done.”

**Why it sticks.** The red team replays every path step by step, and the debrief shows the best plan, so learners immediately want another try.

**Where it fits Ares.** Detection coverage and threat modelling. Credited remix of Threat Modeler by the games team.

WHY PEOPLE REMEMBER IT

# Each game is built on how people learn, not on points and badges.

RETRIEVAL PRACTICE

### The learner produces the answer

They write the SQL, build the rule, place the alert, choose the fix. Producing an answer builds memory that reading and recognising do not.

IMMEDIATE, SPECIFIC FEEDBACK

### Every mistake explains itself

Feedback names the decision and the reason, at the moment it happens. Network Defense alone has about 19 explanation templates.

SAFE FAILURE

### Consequences without damage

A wrong call queues a fraudulent payment or cuts off a working laptop, in fiction. The learner sees the cost, then gets to fix it.

JUDGMENT, NOT RECALL OF FACTS

### Evidence before conclusion

Inspecting never classifies for you. Weak evidence never counts as proof. A correct conclusion without its evidence scores zero in Query Hunt.

REFLECTION

### A debrief that replays your decisions

Choice, observed result, why, and how it connects to the mission. Network Defense rebuilds the exact moment from the command log.

SHORT AND REPEATABLE

### Five minutes, different next time

Seeds and variants change traffic, attackers, the failing link, or which report is benign, so a replay is practice and not memorisation.

One rule runs through all twenty-two: **the game score and the learning evidence are recorded separately.** As one debrief puts it, “this record is evidence, not a grade.” That is what makes the result usable for reporting.

TWENTY YEARS OF GAME CRAFT, APPLIED TO TRAINING

# The hard part of a learning game is balance. I test it the way a game studio would.

If a learner can win by doing the same thing every time, the game teaches that habit. So every game has tests that prove the lazy strategy loses.

| LAZY STRATEGY | WHAT THE TESTS PROVE |
|---|---|
| **Block everything** | “Blanket blocking cannot earn the best result”, checked on 8 seeds |
| **Report every email** | Always-report, always-safe and always-verify each “stay out of the top two ratings” |
| **Escalate everything** | “Penalizes escalating everything alarming” and “penalizes monitoring everything” |
| **Trust the label** | “Never treats the severity tag as a valid justification” |
| **Always investigate first** | “Investigating first is not always right: delay lets the incident progress” |
| **Isolate, flag or bet on everything** | Newer games too: isolate-all stays under 60, flagging everything can’t reach 100, and “certain” on everything loses |
| **Go fast** | “Slow but correct play still earns the top rating”; speed only counts after a correct decision |

### The first ten seconds

A guided, untimed first step in each game. The learner acts before they read a manual, the way a good game tutorial works.

### An economy that forces trade-offs

Action points, analyst slots, credits and case minutes are all limited. Limited resources are what turn knowledge into decisions.

### Feedback you can feel

Routes light up, tokens bounce off a rule, a service goes dark. Shape and icon back up colour, so the signal never depends on colour alone.

### A reason to play again

Ratings, a clean-run target, and a variant that changes the right answer. Query Hunt’s second dataset makes the highest count the wrong account to follow.

TEST NAMES QUOTED FROM THE PROJECT’S TEST SUITES · 318 OF 318 PASSING

BUILT LIKE A PRODUCT, NOT A DEMO

# These are engineered to be reviewed, tested, embedded and extended.

### Content lives in one file per game

Scenarios, evidence, policy and explanations sit in a single data file, apart from the code. A content developer or SME can review and change the teaching without touching the engine.

### The rules run without a browser

Each game's simulation is pure logic, separate from the screen. That is why 318 tests can check the learning rules in under two seconds.

### Deterministic and replayable

The same seed and the same commands always give the same result. It powers the debrief replays, and it would let the platform verify a result on the server.

### A host contract already written

Each game posts started, statement, completed and navigate messages to its host page, with an activity ID per scenario. A draft integration document describes the payloads.

### SME review packs ready to sign

Four games come with a review sheet, and the fourteen newer ones with a design doc ending in an SME checklist. Each lists every objective, simplification and explanation for an expert to approve, correct or reject.

### Accessible by design

Keyboard play, focus-trapped dialogs, screen-reader announcements, and reduced-motion and larger-text settings. Layouts stack for small screens.

**~30,000**
lines of TypeScript, including tests

**22 pages**
one static page per game, one build

**0 servers**
no backend, no range, no sign-in

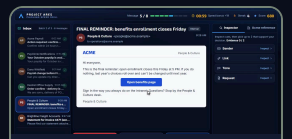
**AI-assisted**
I design and direct, and build with AI tooling. That is how one person ships this much.

# Twenty-two games beside the hub's six.

**The original eight.** Judgment under pressure: incomplete evidence, limited time and a business to keep running.



### Network Defense
Routes, evidence, scoped access rules



### Phishing Speed Drill
Sender, link and reply-to evidence




### Incident Triage Race
Priority from evidence and impact



### SOC Dashboard Challenge
Discriminating evidence, proportionate response



### Cyber Dispatch
First response and clean handoff



### Query Hunt
SQL log investigation

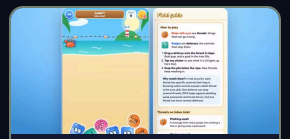


### Cloud Command
Capacity, cost, least privilege

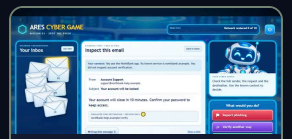



### Privilege Path Puzzle
Groups, grants and minimal fixes

**Eight more, built since.** Scoping, layering, telemetry, subnets, firewall order, awareness.



### Secure Stack
Physics sticker pile



### Spot the Phish
Inbox with a coach




### Blast Radius
Deduction puzzle




### Defense in Depth
Defence planning board




### Counterplay
Card duel




### Bit by Bit
Gear workbench




### Rule Order
Rule-ordering puzzle




### Spot the Risks
Hidden-object scene

**Six remixes.** New twists on the six games built by the games team, each credited to its original.




### Pattern Patrol
Remix of RegExile




### Port Authority
Remix of Portflow




### Confidence Board
Remix of Trivialoot




### Chain Breaker
Remix of Cylitaire: Kill Chain




### Tactic Orbit
Remix of Astro-Defense




### Blind Spots
Remix of Threat Modeler

WHERE IT FITS ARES

# These add to the Games Hub. They cover ground the platform has already marked as open.

The hub's six games cover ATT&CK tactics, the kill chain, regex, ports, trivia and attack paths. The design guide lists what is still uncovered, and the router holds five empty reserved slots.

| OPEN IN THE DESIGN GUIDE OR ROUTER | WHAT I HAVE PLAYABLE |
|---|---|
| Reserved slot `phish-sprint` | **Phishing Speed Drill** |
| Reserved slot `log-hunt` · "Log Analysis / SIEM: entirely untapped" | **Query Hunt**, **Pattern Patrol** |
| Reserved slots `packet-gate`, `network-segmentation-sim` · "Firewall / ACL Rules: entirely untapped" | **Network Defense**, **Rule Order**, **Port Authority** |
| Tier 2 concept "Alert Triage" | **Incident Triage Race**, **SOC Dashboard Challenge** |
| "Cloud Security (IAM, exposure): entirely untapped" | **Cloud Command**, **Privilege Path Puzzle** |
| "Incident Response Lifecycle: entirely untapped" | **Cyber Dispatch**, **Blast Radius**, **Counterplay**, **Chain Breaker** |
| Awareness for all staff (not on the list) | **Secure Stack**, **Spot the Risks**, **Spot the Phish** |

### Different from what exists

The hub's games drill recall of frameworks. These drill judgment: what to do with incomplete evidence, limited time and a business to keep running. The two sets complement each other, and six remixes bridge them: the hub games' topics with a judgment twist.

### They line up with the team's approval gate

The mechanic collapses without domain knowledge. Accuracy outranks speed. Feedback happens during play. Replay is driven by content. Explicit NICE work-role and KSA mapping is the piece still to add, with the content team.

### Small integration step

The hub uses a ready, score-update, complete message flow. My games already post the equivalent events, so each needs a thin adapter, not a rewrite.

GAPS QUOTED FROM DOCS/GUIDES/MINIGAMES (README AND GAME DESIGN GUIDE)

MORE GAMES, IN ANY STYLE

# Twenty-one genres so far. The method is what scales, not one look.

## How a new game gets made

- **Start from the skill.** One learning objective, one decision the learner keeps getting wrong.
- **Find the mechanic that is the skill.** If you can win without the knowledge, it is the wrong mechanic.
- **Write the content as data.** Scenario, evidence, and an explanation for each mistake, in one reviewable file.
- **Prove the balance.** Tests that show the lazy strategies lose and careful play wins.
- **Hand the SME a review sheet.** Every objective and simplification, ready to approve or correct.

The shared pieces already exist: the seeded random generator, the replay log, the host messages, the accessibility helpers and the build. A new game is a data file, a simulation, a screen and a test file.

## What I would build next

| CONCEPT | WHY |
|---|---|
| **IR Sequencer** | Tier 1 in the design guide. Order the response steps under pressure. |
| **CSF Incident Sorter** | Tier 1. Route events to Identify, Protect, Detect, Respond, Recover. |
| **ATT&CK Technique Mapper** | Tier 1. Takes the hub from tactic level to technique level. |
| **More cases for every game** | New scenarios are content, not code. Query Hunt Case 02, a second phishing inbox, a harder shift. |
| **Team modes for workshops** | Head-to-head triage and dispatch for Security Immersion Workshop events, with a shared leaderboard. |
| **Mission warm-ups** | A two-minute game before a range mission that drills the one concept the mission depends on. |

WHAT IT DOES FOR CIRCADENCE

# Games are the cheapest way to add practice, reach and proof to the platform.

### Practice that costs nothing to run

A mission allocates live virtual machines. A game is a static page. Learners can practise as often as they like without a range starting up.

### A demo that runs anywhere

No sign-in and no install. A prospect can play a round in a meeting, on a booth screen, or from a link in an email, and understand what Ares teaches.

### More for every licence

New games widen the catalog between major releases, and give customer admins something fresh to assign.

### Content for workshops and events

Short, scored, spectator-friendly games suit Security Immersion Workshops, where dozens of people play at once from a browser.

### Evidence for reporting

Each game records decisions and reasoning separately from the score. Once mapped to work roles, that feeds readiness reporting with more than a completion tick.

### Ready for the Japanese market

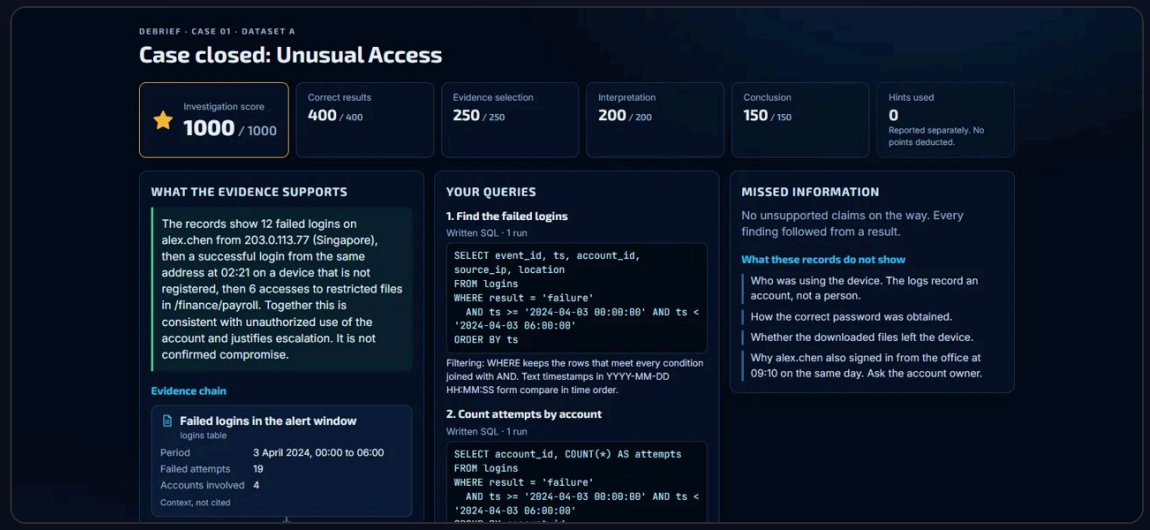
Text is held as data, so games can be localised the same way as the platform. I have already worked on the Japanese strings across the player app.

I have not measured learner outcomes yet, and I am not claiming numbers. The first thing I would do with the team is put these in front of real learners and measure completion, replays and what they get wrong.

HONEST STATUS

# They are prototypes. Here is exactly what stands between them and production.

| OPEN ITEM | WHAT IT NEEDS |
|---|---|
| **SME review** | All content is draft. Four games have review sheets and fourteen have SME checklists; four still need one. I need a few hours of expert time per game. |
| **Learner playtesting** | None yet. Internal staff first, then a friendly customer cohort. |
| **Framework mapping** | The games do not yet name NICE work roles or KSAs. That mapping should be done with the content team so it matches the catalog. |
| **Platform integration** | The host contract is a draft that has not been agreed with the platform team. Course and mission text is mocked. |
| **Localisation** | English only today. Japanese follows SME sign-off. |
| **Borrowed art and remixes** | Network Defense reuses sprites from another developer, with their go-ahead; I would confirm in writing or replace them. The six remixes credit the games team and need its blessing before they ship. |
| **Depth** | Three games have seeded variation, two have a second variant, three have one scenario. More cases are content work. |




## Why I am showing you the gaps

Because closing them is the job I am asking to do. None of it is research. It is review, integration and content, and I know exactly what each game needs.

THE FIRST 90 DAYS

# From twenty-two prototypes to games in the hub, with the team.

DAYS 1 TO 30

## Get three in

- Agree the host contract with the platform team and write the hub adapter.
- SME review of Network Defense, Query Hunt and Cloud Command, using the sheets that exist.
- Map those three to NICE work roles and KSAs with the content team.
- Confirm or replace the borrowed art.

DAYS 31 TO 60

## Test and widen

- Internal playtest. Measure completion, replays and common mistakes, then fix what confuses people.
- Review sheets and SME pass for Phishing, Triage and SOC.
- Japanese strings for the first three.
- A second case for the game learners replay most.

DAYS 61 TO 90

## Finish and extend

- Dispatch and Privilege Path through review and into the hub.
- First new game from the design guide’s Tier 1 list, built with the games team.
- An authoring guide so content developers can add scenarios themselves.
- A short report on what the playtests showed.

These are targets, and they depend on SME and platform time I do not control. I would rather agree the order with you than promise a date I cannot own.

WHAT IS ALREADY ON THE TABLE

# This is working software, not a proposal. Starting again would mean rebuilding five things at once.

| PIECE | WHAT EXISTS TODAY | WITHOUT IT |
|---|---|---|
| **Game design** | Twenty-two games on twenty-one mechanics, each one built so it cannot be won without the knowledge | Finding a designer who knows both games and cyber training |
| **Authored content** | 8 phishing messages with 32 cues, 6 alerts with 29 evidence items, 6 dispatch reports with 38 questions, 5 SQL challenges on 2 datasets, 3 access puzzles, 3 cloud checkpoints, and an explanation for every important mistake | Writer and SME time for every scenario from a blank page |
| **Engineering** | About 30,000 lines of TypeScript, twenty-one static game pages, one shared pattern | A front-end build from an empty folder |
| **Balance and proof** | 318 passing tests, including the ones that prove the lazy strategies lose | Finding the exploits the hard way, with real learners |
| **Review and integration** | Four SME review packs ready to sign, and a draft host contract with an activity ID per game | Starting both conversations from nothing |

### Ready to show a customer

Any of the twenty-two can be demoed in a browser today. No range to start, no sign-in, nothing to install. That is a new thing to put in front of a prospect this quarter.

### The rare part is the combination

Game design, UX, front-end build, and knowledge of Ares from the inside. My 89 commits in the platform cover the world map, dashboard, debrief and the HUD screens, so these games are built to sit in the product, not beside it.

### One owner, low cost

This does not need a new team or new infrastructure. It needs the person who built it, some SME hours, and a small integration step.

COUNTS TAKEN FROM THE GAMES' DATA FILES, TEST SUITES AND DOCS

WAYS TO KEEP THIS GOING

# I want to stay and build this with you. Here are three ways to make it work.

PREFERRED

## Full time, games and learner experience

**Current hours and rate**

I move toward the games team. The 90-day plan runs as written, and I keep supporting the player UI I already own: world map, dashboard, debrief and the HUD screens.

REDUCED

## Half time, games only

**About 20 hours a week**

One focus: take the games to production in priority order, then new games. The plan stretches to roughly twice the calendar time. UI work is by request.

MINIMUM

## Per game, fixed scope

**Agreed game by game**

You pick a game. I deliver it reviewed, integrated and tested for an agreed scope. No standing commitment, and the work continues at whatever pace you choose.

### What you keep

Twenty-two working games and the person who knows every rule in them. A designer with twenty years of game experience who also builds. Someone already inside the product: the world map, dashboard, debrief and 16 HUD screens are my work.

### What stops without an owner

These games are not in the platform repository yet. Without someone to take them through review and integration, they stay prototypes, and the open slots in the hub stay open.

**My ask:** thirty minutes to play two of them with you, and a decision on which option fits. The games are ready to demo to a customer now; the only open question is who takes them the rest of the way.